THE EXTENDED ALTERNATING CECH COMPLEX OF A KOSZUL REGULAR SEQUENCE IS CONCENTRATED IN THE TOP DEGREE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Products of a Koszul regular sequence with a repeated element remain Koszul regular. Lemma \ref{lemma-mult-koszul-regular}, cited at proof lines 7873-7874. Lets the last element be replaced by any power.
- A permutation of a Koszul regular sequence is Koszul regular. Lemma \ref{lemma-change-basis}, cited at proof lines 7875-7876. Lets any element be replaced by a power.
- The extended alternating Cech complex is a colimit of Koszul complexes. Lemma \ref{lemma-extended-alternating-Cech-is-colimit-koszul}, cited at proof lines 7880-7881. Transfers the concentration of the Koszul cohomology to the Cech complex.

Source reference. Original lemma 0G6L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20201–20201. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0G6F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20195–20195. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 062G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7795–7795. Navigation pointer to a statement the

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Source reference. Original lemma 0625: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7784–7784. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0913: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11431–11431. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let $f_1, \ldots, f_r \in R$ be an Koszul-regular sequence. Then the extended alternating Čech complex $R \to \bigoplus_{i_0} R_{f_{i_0}} \to \bigoplus_{i_0<i_1} R_{f_{i_0}f_{i_1}} \to \ldots \to R_{f_1\ldots f_r}$ from Section 0G6F only has cohomology in degree r.*

Proof. By Lemma 062G and induction the sequence $f_1, \ldots, f_{r-1}, f_r^n$ is Koszul regular for all $n \geq 1$. By Lemma 0625 any permutation of a Koszul regular sequence is a Koszul regular sequence. Hence we see that we may replace any (or all) f_i by its nth power and still have a Koszul regular sequence. Thus $K_\bullet(R, f_1^n, \ldots, f_r^n)$ has nonzero cohomology only in homological degree 0. This implies what we want by Lemma 0913. □